Hints: Limits – 3. Asymptotics of sequences

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The map $g(x) = x - \arcsin(\sin^3 x)$ is odd, so it suffices to take $x_0 \in (0, 1]$ (for $x_0 = 0$ the sequence is 0). For $x \in (0, 1]$: $0 < \sin^3 x < \sin x$, so $0 < \arcsin(\sin^3 x) < x$; hence x_n stays in $(0, 1]$, decreases, and tends to a fixed point, i.e. to 0.

Hint 2. Here $\arcsin(\sin^3 x) = x^3 + O(x^5)$, so $x_{n+1} = x_n - x_n^3 + O(x_n^5)$: Theorem 1 with $p = 3$, $a = 1$. Telescoping: $\dfrac{1}{x_{n+1}^2} - \dfrac{1}{x_n^2} \to 2$, and Stolz gives $nx_n^2 \to \frac{1}{2}$.

Hint 3. Answer: $\lim \sqrt{n}\, x_n = \frac{1}{\sqrt{2}}$ for $x_0 \in (0, 1]$, $-\frac{1}{\sqrt{2}}$ for $x_0 \in [-1, 0)$, and 0 for $x_0 = 0$.

Problem 2

Hint 1. Substitute $t = nx$: $b_n = \frac{1}{n}\int_0^n f(t)\,dt = \frac{1}{n}\sum_{k=0}^{n-1} a_k$.

Hint 2. So b_n is the Cesàro mean of (a_k), and converges to the same limit (Stolz–Cesàro).

Problem 3

Hint 1. The recurrence is $\frac{x_{n+1}}{x_n} = 2018 \cdot \frac{n+1}{n}$; multiply these ratios.

Hint 2. $x_n = 2018^n\, n$, so $n - \log_{2018} x_n = -\log_{2018} n$, and the limit is $\dfrac{\log_2 n}{-\log_{2018} n} = -\log_2 2018$.

Problem 4

Hint 1. Compute the first terms: $1, 3, 2, \frac{3}{2}, \frac{7}{4}, \frac{13}{8}, \frac{25}{16}, \frac{51}{32}, \ldots$ and keep track of which of the three previous terms is discarded (the largest).

Hint 2. From (b_5, b_6, b_7) on, the pattern is periodic with period 3: if $p > q > r$ are three consecutive terms, the next three are $s = \frac{q+r}{2}$, $t = \frac{q+3r}{4}$, $u = \frac{q+7r}{8}$, and again $s > t > u$. So the pair (q, r) is mapped to (t, u), independent of p.

Hint 3. The map $(q, r) \mapsto (\frac{q+3r}{4}, \frac{q+7r}{8})$ preserves $q + 6r$ and divides $q - r$ by 8. Starting from $(b_6, b_7) = (\frac{13}{8}, \frac{25}{16})$, $q + 6r = 11$, so $\lim b_n = \frac{11}{7}$.

Problem 5

Hint 1. The candidate is $L = 4$ ($L = 2\sqrt{L}$). Note c_{n+1} lies between $2\sqrt{\min(c_n, c_{n-1})}$ and $2\sqrt{\max(c_n, c_{n-1})}$, and $2\sqrt{x} \geqslant x$ exactly for $x \leqslant 4$.

Hint 2. Hence with $m_n = \min(c_n, c_{n-1}, 4)$ and $M_n = \max(c_n, c_{n-1}, 4)$ we get $c_{n+1} \in [m_n, M_n]$: the sequence is bounded and bounded away from 0.

Hint 3. Let $M = \limsup c_n$, $m = \liminf c_n$ (both in $(0, \infty)$). Passing to the limit, $M \leqslant 2\sqrt{M}$ and $m \geqslant 2\sqrt{m}$, so $M \leqslant 4 \leqslant m$; thus $c_n \to 4$.

Problem 6

Hint 1. Rewrite: $x_{n+2} - 2x_{n+1} = (x_{n+1} - 2x_n) + \frac{2^n}{x_n}$. So $d_n = x_{n+1} - 2x_n$ satisfies $d_1 = 0$, $d_{n+1} = d_n + \frac{2^n}{x_n}$.

Hint 2. Put $y_n = x_n/2^n$. Dividing by 2^{n+1}: $y_{n+1} - y_n = \frac{d_n}{2^{n+1}}$, and $\frac{2^n}{x_n} = \frac{1}{y_n}$, so $d_n = \sum_{k=1}^{n-1} \frac{1}{y_k}$. Hence $d_n \geqslant 0$, y_n is non-decreasing, $y_n \geqslant y_1 = 1$ and $d_n \leqslant n-1$.

Hint 3. Then $0 \leqslant y_{n+1} - y_n \leqslant \frac{n-1}{2^{n+1}}$ is summable, so $L = \lim y_n$ exists and $L \leqslant 1 + \sum_{n\geqslant 1} \frac{n-1}{2^{n+1}} = \frac{3}{2}$. Using $y_k \leqslant \frac{3}{2}$ in d_n gives $L \geqslant 1 + \frac{2}{3} \cdot \frac{1}{2} = \frac{4}{3}$.

Hint 4. Iterating these bounds (compute the first few y_k exactly, estimate the tail) sharpens them; numerically $L \approx 1.4782$.

Problem 7

Hint 1. Factor: $(2n)^4 - 1 = (1 - \frac{1}{(2n)^2})(1 + \frac{1}{(2n)^2})(2n)^4$ and similarly for $(2n+1)^4 - 1$. Then the product splits into products whose values come from the sine product $\frac{\sin \pi x}{\pi x} = \prod_{n\geqslant 1}(1 - \frac{x^2}{n^2})$ (and its versions for cos, sinh, cosh).

Hint 2. Needed: $\prod_{n\geqslant 1}(1 - \frac{1}{4n^2}) = \frac{2}{\pi}$, $\prod_{n\geqslant 1}(1 + \frac{1}{4n^2}) = \frac{2\sinh(\pi/2)}{\pi}$, $\prod_{n\geqslant 1}(1 - \frac{1}{(2n+1)^2}) = \frac{\pi}{4}$, $\prod_{n\geqslant 1}(1 + \frac{1}{(2n+1)^2}) = \frac{\cosh(\pi/2)}{2}$. The remaining factor is $\frac{(2n)^4(n+1)^2}{(2n+1)^4 n^2} = (1 - \frac{1}{(2n+1)^2})^2$.

Hint 3. Combining, the product equals $\dfrac{2}{\pi} \tanh \dfrac{\pi}{2} \approx 0.5839$.

Problem 8

Hint 1. Invert: $\dfrac{1}{a_{n+1}} = \dfrac{1}{a_n} + 1 + a_n$.

Hint 2. Hence $\frac{1}{a_n} \geqslant \frac{1}{a_0} + n$, so $a_n \to 0$; then $\frac{1}{a_{n+1}} - \frac{1}{a_n} \to 1$ and Stolz gives $\lim n a_n = 1$.

Problem 9

Hint 1. $x_n = P_n(x_1)$, where P_n is a polynomial with non-negative coefficients and no constant term; on $(0, \infty)$ it is strictly increasing. Note $x_{n+1} > x_n \iff x_n > 1 - \frac{1}{n}$.

Hint 2. So the condition is: $x_1 > 0$ and $1 - \frac{1}{n} < P_n(x_1) < 1$ for all n. Let $a_n, b_n > 0$ be defined by $P_n(a_n) = 1 - \frac{1}{n}$, $P_n(b_n) = 1$. Then $P_{n+1}(a_n) = (1 - \frac{1}{n}) \cdot 1 < 1 - \frac{1}{n+1}$ and $P_{n+1}(b_n) = 1 + \frac{1}{n} > 1$, so a_n increases and b_n decreases; nested intervals give existence.

Hint 3. Uniqueness: $P_n(x)/x$ is non-decreasing, so $P_n(b_n) - P_n(a_n) \geqslant (b_n - a_n)\frac{P_n(a_n)}{a_n}$, i.e. $\frac{1}{n} \geqslant (b_n - a_n)\frac{1 - 1/n}{a_n}$. With $a_n < 1$ this gives $b_n - a_n \leqslant \frac{1}{n-1} \to 0$.

Problem 10

Hint 1. Fix $\varepsilon > 0$ and let $E_N = \{x > 0 : |f(nx)| \leqslant \varepsilon \text{ for all } n \geqslant N\}$. Each E_N is closed in $(0, \infty)$ (continuity of f), and $\bigcup_N E_N = (0, \infty)$ by hypothesis.

Hint 2. By Baire's theorem some E_N contains an interval (a, b), $0 < a < b$. Then $|f| \leqslant \varepsilon$ on $\bigcup_{n\geqslant N}(na, nb)$.

Hint 3. For $n > \frac{a}{b-a}$ consecutive intervals (na, nb) and $((n+1)a, (n+1)b)$ overlap, so this union contains a half-line (M, ∞). Hence $|f(x)| \leqslant \varepsilon$ for $x > M$.